



Similarity-Promoting Resilient Adaptive Aggregation in Peer-to-Peer Machine Learning

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Abstract

Peer-to-peer collaborative learning offers a distributed approach that eliminates the risk of single-point-of-failures seen in federated learning. However, the presence of adversarial workers in the network, who aim to spread malicious information, presents a significant threat. Thus, ensuring the resilience of peer-to-peer learning emerges as a pivotal research objective. The challenge of safeguarding peer-to-peer learning against malicious attacks becomes even more difficult with non-convex loss functions and non-iid data distributions among the workers. This paper focuses on ensuring the robustness of distributed peer-to-peer learning in non-convex loss and non-iid data distribution scenario by using a resilient aggregation technique. This novel aggregation is designed by fostering similarity among the workers' learning behaviors. Each worker aggregates the parameters of its neighbors as a weighted sum, where the weights are determined through an optimization problem aimed at learning the optimal parameters of each worker while protecting data privacy. The workers compute the loss using their neighbor's models and individual private data, thus protecting data privacy. Theoretical analysis demonstrate that the normal workers' parameters reach a consensus. Additionally, the difference between the parameters of normal workers and their respective optimal values remains bounded, with the bound being a function of few hyperparameters and the variance of non-iid data distribution among the workers. The effectiveness of the approach is evidenced by empirical evaluations performed on three machine learning classification tasks. The results demonstrate that the proposed adaptive aggregation method improves the test accuracy of the normal workers when compared to state-of-the-art aggregation techniques, under various adversarial scenarios spanning multiple attack models.

Keywords Distributed machine learning · Resilient learning · Peer-to-peer learning · Adaptive aggregation · Non-convex loss function · Non-iid data distribution

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1 Introduction

Machine learning (ML) continues to evolve in various sectors of modern-day life, thriving due to advanced hardware and improved sensors. Distributed ML capitalizes on the vast computational resources and diverse datasets spread across multiple nodes, as found in sectors such as Internet-of-Things (IoT) systems (Nguyen et al., 2021), sensor networks (Seng et al., 2022), smart city (Jiang et al., 2020), autonomous vehicles (Nguyen et al., 2022; Zhang et al., 2021), recommender systems (Cui et al., 2020) and many more. It is capable of processing large amount of data much faster than centralized systems and fosters collaborative knowledge discovery among the nodes. With growing concerns of data privacy in application areas such as healthcare or finance, distributed ML is able to train ML models across multiple worker machines without them having to share data. Federated learning (FL) became popular due to its ability to train a ML model by using multiple workers, coordinated by a centralized parameter server (McMahan et al., 2017; Bonawitz et al., 2017).

Peer-to-peer (P2P) learning networks stand as a compelling alternative to traditional federated learning (FL) by promoting a decentralized model of training, which inherently provides resilience against the single points of failure often found in FL architectures. In P2P learning, each *worker*, or *peer*, acts both as a client contributing to the learning process, and as a server by collecting the insights of other peers, thus eliminating the need for a central coordinating server. This not only distributes the computational load but also democratizes the learning process, as every peer has equal standing and can directly exchange information with others. By doing so, P2P networks offer improved fault tolerance and potentially enhanced privacy, as data can be kept local while still contributing to the collective learning goal. Additionally, P2P systems offer more flexibility to dynamic scenarios where peers can be added or removed easily without disrupting the learning process. This advantage is well-suited for scenarios with intermittent connectivity or in ad-hoc networks. The inherent scalability and efficiency of P2P learning make it an attractive paradigm for handling large-scale, distributed datasets in a more secure and resilient manner.

Despite these advantages, P2P learning faces some unique challenges. In the realm of security, the absence of a central server creates vulnerabilities, making the system more susceptible to attacks by adversarial peers that may enter the network with the intent of disrupting the learning process. These adversarial entities can inject falsified data or share adversarial parameters with their neighbors. Such threats require robust defense mechanisms to detect and mitigate such attacks, without which the integrity of overall learning could be compromised. Furthermore, P2P learning has inherent challenges - the distributed data across workers may not be identically and independently distributed (non-IID), which poses additional complexities in ensuring the effectiveness of the learned models when naive communication and aggregation are permitted. Another inherent challenge of P2P learning is the non-convex loss function. Ensuring resilience in P2P learning becomes more difficult in the presence of these challenges.

While resilient P2P learning has garnered significant research attention, there is a noticeable gap in the literature where very few works address all three critical challenges mentioned above. Most existing algorithms focus only on addressing a subset of these issues. For example, several resilient algorithms are restricted to convex loss functions. Notably, the ByRDIE algorithm (Yang & Bajwa, 2019) that employs a decentralized coordinate descent-based learning strategy and MOZI (Guo et al., 2021) that utilizes a two-step filter-

ing mechanism - both achieve robustness against Byzantine adversaries but are limited to handling convex loss functions. Other works, such as Peng et al. (2021), extend resilience to more complex settings involving non-uniformly distributed data and time-varying networks. However, their methodology does not comprehensively address the challenge posed by non-convex loss landscapes. While some studies have attempted to relax the convexity assumption, limitations remain. The work in Fang et al. (2022) developed resilient P2P algorithms for both convex and non-convex loss functions, but requires the data distribution to be iid.

There is a growing concern and an urgent need to develop resilient P2P learning algorithms capable of handling non-uniform data distributions and non-convex loss functions. While some efforts have been made to address these challenges, limitations remain. For instance, the work in He et al. (2022) extends its applicability to a broader class of loss functions and non-iid data distributions. However, it relies on restrictive assumptions, such as normal workers knowing the identities of attacked workers and the negligible influence of adversarial gradients, which may not hold in practical scenarios. Similarly, the ICwTM algorithm proposed in El-Mhamdi et al. (2021) requires the exchange of both local gradients and local models, leading to increased communication overhead. Additionally, it imposes the constraint that all workers must share the same initialization. Both approaches also maintain a restrictive upper bound on the number of tolerable adversarial workers, which motivates this work to improve resilience while relaxing these limitations.

In this paper, we develop a distributed P2P learning algorithm capable of tolerating more Byzantine workers than any state-of-the-art methods, and is functional in non-iid distribution of data and non-convex loss function scenarios. Considering some workers are adversarial and share malicious information with their neighbors, we propose a novel loss-based adaptive aggregation method to ensure robustness of the normal workers. This method provides resilience in the presence of an arbitrary number of adversarial neighbors under the graph-redundancy condition stated in Assumption 5, and works by promoting similarity between the workers' learning behavior. Each peer aggregates the parameters of its neighbors by computing the loss using its own data and neighbors' parameters. A smaller loss indicates that their objectives are related, so a higher aggregation weight is assigned. On the other hand, an adversarial neighbor inherently possesses a different objective than the worker itself, and thus it receives a smaller aggregation weight. This idea has been proven effective in context of multi-task learning (Li et al., 2020) and multi-agent reinforcement learning (Bhowmick et al., 2023). This method is shown effective for a broad range of model poisoning attacks.

This paper presents the following key contributions:

- We propose a novel adaptive parameter aggregation technique for collaborative peer-to-peer (P2P) learning, where the workers have non-i.i.d. local datasets and optimize non-convex loss functions. In this privacy-protected learning setup, each worker aggregates parameters shared by its neighbors as an adaptive sum. The proposed aggregation method promotes similarity between workers' learning, by computing the loss using worker's private data and parameters shared by the neighbors. Aggregation weights are determined as functions of these computed losses, by solving an optimization problem that aims to make the worker parameters converge to their optimal values.
- We present theoretical analysis to demonstrate consensus and optimality of the learned

parameters of the normal workers. In the consensus or agreement analysis, we show that the normal workers' parameters asymptotically converge to the same value. Additionally, we prove optimality of converged parameters, showing that the normal workers' parameters converge close to their respective optimal values with a derived upper bound, which is a function of the type of loss function, learning rate and the distribution of data among the workers. These analyses demonstrate the learning behavior of the normal workers under the graph-redundancy condition in Assumption 5, in the presence of an arbitrary number of adversarial workers in their neighborhood - showing that the proposed adaptive aggregation method is resilient to more adversarial workers as compared to any existing methods in the literature, when the communication graph provides sufficient redundancy.

- A detailed empirical evaluation is included that considers three ML classification tasks with non-convex loss function and non-iid data distribution across the workers. The similarity-promoting adaptive aggregation demonstrates improved performance in terms of accuracy and resilience compared to most state-of-the-art P2P learning methods, under a variety of poisoning attacks. For instance, in human activity recognition task under arbitrary Byzantine or time-coupled attacks, our method achieves more than 94% accuracy, resulting an absolute improvement of at least 74 percentage points over the best baseline ($< 20\%$).

The rest of the paper is organized as follows. Relevant works from the literature are mentioned in Sect. 2. Section 3 provides a background on peer-to-peer learning, possible poisoning attacks on them and existing resilient aggregation techniques. Section 4 presents the similarity-promoting loss-based adaptive aggregation technique. The theoretical analysis including consensus analysis and convergence toward optimality are included in Sect. 5. Empirical results and discussions are provided in Sect. 6. Finally, Sect. 7 concludes the paper.

2 Related Work

2.1 Applications of Distributed ML

Distributed learning has a wide range of applications across various domains due to its ability to train machine learning models collaboratively while preserving data privacy and decentralizing the training process (Li et al., 2020a, b). The key advantages of P2P learning—*scalability, robustness, and privacy preservation*—have made it applicable across diverse domains (Vanhaesebrouck et al., 2017; Bellet et al., 2018; Elgabli et al., 2020; Colin et al., 2016; Lalitha et al., 2019; Tang et al., 2018; Nedić et al., 2018).

In recommender systems, P2P learning supports personalized recommendations without requiring centralized access to sensitive user-item interactions. Several studies demonstrate its ability to enhance personalization while reducing privacy-leakage risks associated with sparse or user-specific data (Wu et al., 2021; Ivannikova et al., 2019; Lin et al., 2020; Liang et al., 2021).

In the healthcare domain, decentralized P2P frameworks enable hospitals and edge devices such as wearables to collaboratively train models on sensitive medical or physi-

ological data without sharing raw records. This enables applications such as patient monitoring, electronic health record mining, and drug discovery while maintaining strict privacy guarantees (Jiang et al., 2020; Mozaffari-Kermani et al., 2014; Terrail et al., 2022).

For cyber-physical and IoT systems, P2P learning leverages large networks of connected sensors and devices to perform tasks such as resource allocation, anomaly detection, and adaptive control in smart grids, autonomous vehicle systems, and industrial automation. These networks benefit from P2P learning's resilience to faults and adversarial agents, as no single point of failure exists (Samarakoon et al., 2018; El Hanjri et al., 2023; Al-Huthaifi et al., 2023; Abou El Houda et al., 2023).

In finance, P2P learning enables risk prediction and fraud detection across institutions without exposing sensitive transaction data (WeBank, 2019; Schreyer et al., 2022). Similarly, in *smart-city* infrastructures, peer-to-peer coordination among distributed agents facilitates traffic optimization, energy management, and intrusion detection in a fully decentralized manner (Chen et al., 2021; Pokhrel & Choi, 2020).

Finally, the foundations of P2P learning are closely connected to the literature on multi-agent optimization and control, where consensus, gossip, and distributed gradient methods have long served as coordination mechanisms in multi-agent systems and now underpin decentralized ML frameworks operating over dynamic, large-scale networks (Nedić et al., 2018).

2.2 Peer-to-Peer Learning Algorithms

P2P learning eliminates the central server by leveraging direct information exchange between workers (Yuan et al., 2024). The communication often relies on *gossip protocols* (Xiao & Boyd, 2004), where nodes iteratively average their local models with those of randomly chosen neighbors, ensuring information diffusion across the network. The choice of *topology* for the communication graph of workers plays a central role: dense and well-connected graphs accelerate convergence, while sparse or time-varying topologies introduce challenges in mixing rates and stability (Nedić et al., 2018).

A foundational P2P method is Decentralized Parallel Stochastic Gradient Descent (D-PSGD) (Lian et al., 2017), which combines stochastic gradient descent (SGD) with gossip-based averaging. Building on this, Decentralized Momentum SGD (DMSGD) introduced momentum to accelerate convergence (Balu et al., 2021). Other extensions generalize D-PSGD to heterogeneous settings and non-convex problems (Koloskova et al., 2020; Assran et al., 2019).

More broadly, SGD has proven powerful for solving large-scale ML problems (Ghadimi & Lan, 2013; Moulines & Bach, 2011) and underpins many decentralized architectures. Early works such as Colin et al. (2016) proposed decentralized algorithms for pairwise function optimization. Variants of Alternating Direction Method of Multipliers (ADMM) have also been adapted to decentralized optimization (Mokhtari & Ribeiro, 2016; Shi et al., 2014). A challenge in these methods is the step-size requirement. Some algorithms guarantee convergence only with diminishing step sizes (Nedic & Ozdaglar, 2009). Others such as EXTRA (Shi et al., 2015a) and proximal-gradient methods (Shi et al., 2015b) converge under constant step sizes determined by the Lipschitz constant and the network topology. Later developments (Li et al., 2019; Yuan et al., 2018) introduced algorithms with network-independent constant step-sizes, improving scalability.

Beyond SGD, dual-averaging approaches (Duchi et al., 2011) and convergence rate analyses (Yuan et al., 2016) have enriched the theoretical foundation of decentralized optimization. Importantly, Lian et al. (2017) showed that decentralized parallel learning scales with the number of workers, achieving convergence rates comparable to centralized methods.

Recent P2P learning research has also emphasized communication efficiency and personalization in decentralized settings. GossipFL (Tang et al., 2022) proposes a gossip-based decentralized federated learning framework with sparsified and adaptive communication to reduce bandwidth usage. Decentralized Directed Collaboration (Liu et al., 2024) introduces a directed collaboration graph to enable personalized model updates among selected peers. Similarly, DispFL (Dai et al., 2022) leverages decentralized sparse training to improve communication efficiency and personalization, while Hot-Pluggable Federated Learning (Shen et al., 2025) dynamically selects collaborators to bridge general and personalized models. These approaches primarily focus on communication efficiency and personalization under heterogeneous but benign settings, and do not explicitly address robustness to adversarial or Byzantine model updates.

Most theoretical results in P2P learning assume bounded variance across worker data (Lian et al., 2018, 2017; Nedic & Ozdaglar, 2009; Yuan et al., 2016). While variance-reduction methods were first studied in centralized settings (Konečný et al., 2016; Schmidt et al., 2017), adaptations to decentralized optimization have also been developed. For example, Mokhtari and Ribeiro (2016) applied variance reduction techniques to strongly convex problems, and Tang et al. (2018) proposed a decentralized algorithm robust against high variance among workers.

2.3 Attacks on Distributed Machine Learning

In general, attacks can be categorized as *poisoning* or training-phase attacks and *evasion* or testing-phase attacks (Bhagoji et al., 2019; Liu et al., 2018). Poisoning attacks can focus on *data poisoning* (Biggio et al., 2012; Liu et al., 2018), and *model poisoning* (Bagdasaryan et al., 2020; Bhagoji et al., 2019). Data poisoning attacks include *label flipping* and *backdoor insertion* (Chen et al., 2017). Model poisoning attacks, which is the focus of this paper, target the information shared by the agents, and include *gradient manipulation* (Bagdasaryan et al., 2020) and *training rule manipulation* (Bhagoji et al., 2019). Moreover, poisoning attacks can be designed given a specific objective by solving an optimization problem as, for example, in Muñoz-González et al. (2017). A class of state-of-the-art model poisoning attacks, called the time-coupled attacks, operate by introducing slight perturbations to the model, resulting in the aggregated model deviating significantly from the correct model in course of time. *A-little-is-enough (ALIE)* (Baruch et al., 2019) is a non-omniscient time-coupled attack reliant on minor perturbations in the attack vector. Another example is the *Fall-of-empire (FoE) attack* (Xie et al., 2020), which manipulates the inner-product between the true and the aggregated parameters.

2.4 Resilient Distributed Machine Learning

One of the earliest works in resilient P2P is ByrDiE (Yang & Bajwa, 2019) that employs a decentralized coordinate-descent-based learning strategy to ensure Byzantine fault tolerance. A fully decentralized P2P learning algorithm using blockchain and cryptographic

approaches was proposed in Biscotti (Shayan et al., 2020). This method is shown to be resilient to up to 30% adversaries, scalable, and protective of data privacy. In another study, El-Mhamdi et al. (2021) proposed a decentralized learning paradigm suitable for heterogeneous, asynchronous, and nonconvex settings that is resilient against Byzantine workers when the condition $N \geq 6F + 1$ holds, where N and F are the total number of workers and the number of Byzantine workers, respectively. In BRIDGE (Fang et al., 2022), multiple aggregation techniques such as coordinate-wise median, trimmed mean, Krum, and Bulyan were explored to design Byzantine-resilient P2P learning algorithms. MoNNA (Farhadkhani et al., 2023) is an algorithm that uses Polyak's momentum of local gradients for local updates and nearest-neighbor averaging (NNA) for the aggregation of neighbors' information, and is suitable for asynchronous communication. Most of these works operate on strong assumptions such as homogeneous data (Fang et al., 2022; Gorbunov et al., 2022; Guo et al., 2021; Yang & Bajwa, 2019), or strong convexity (Gupta et al., 2021; Yang & Bajwa, 2019). After centered-clipping aggregation was shown to achieve resilience in a federated setup (He et al., 2022), a recent work (Yang & Ghaderi, 2024) used a remove-then-clip aggregator in a P2P setting. In this setup, each worker filters the neighbors' parameters and projects them onto a sphere centered at its local value, with an appropriately selected radius.

3 Problem Statement

This section discusses operation of peer-to-peer learning, attack models, and resilient aggregation methods.

3.1 Peer-to-Peer Learning

Consider a connected network of N nodes, denoted by a static and undirected graph $\mathcal{G} = \{J, \mathcal{E}\}$, where $J = \{1, 2, \dots, N\}$ is the set of N nodes that represent the workers, and the set of edges $\mathcal{E}$ represents the communication between the workers. Specifically, $(i, j) \in \mathcal{E}$ implies that node i and node j share information with each other. The neighborhood of worker k is the set $\mathcal{N}_k = k \cup \{l \in J | (l, k) \in \mathcal{E}\}$.

These workers have access to private data, and they collaborate with each other to solve a supervised ML problem in a distributed manner. The local training dataset of worker k is given by $S_k = \{\xi_k^i\}_{i=1}^{|S_k|}$, where $\xi_k^i = (x_k^i, y_k^i)$ is a data sample of worker k , with $x_k^i \in \mathbb{R}^{d_x}$, $y_k^i \in \mathbb{R}^{d_y}$. We assume that the distribution of data among these workers is non independent and identically distributed (non-iid).

We use $\ell(w_k; \xi_k)$ to denote a non-negative and possibly regularized loss function that maps (w, ξ) to the corresponding loss $\ell(w, \xi)$. The regularized loss function is often referred to as the risk function $r(\cdot)$. The model w is assumed to be parametric, i.e., $w \in \mathbb{R}^d$. Each worker aims to learn a personalized model by minimizing the local risk function over its local data, while facilitating collaboration with neighbors.

The objective of learning is formulated as:

$$w^* = \min_{\{w_1, w_2, \dots, w_N\}} \frac{1}{N} \sum_{k=1}^N r_k(w_k), \quad (1)$$

where $r_k(w_k) := \frac{1}{|S_k|} \sum_{i=1}^{|S_k|} \ell(w_k; \xi_k^i)$ is the local empirical risk function of worker k . For this work, we assume that the risk function is non-convex.

During each training epoch t , worker k minimizes its local risk using stochastic gradient descent (SGD) on local dataset, using the local update step:

$$\hat{w}_k^t = w_k^{t-1} - \rho^t \cdot \nabla l(w_k^{t-1}; \xi_k^{t-1}), \quad (2)$$

where $\hat{w}_k^t$ is the intermediate updated parameters of worker k and ρ^t is the learning rate of epoch t . Following the local training step, every worker broadcasts the intermediate parameters for their neighbors. Upon receiving the neighbors' parameters, each worker combines them to compute the updated parameters, using the following equation

$$w_k^t = \sum_{l \in \mathcal{N}_k} c_t(l, k) \cdot \hat{w}_l^t$$

(3)

subject to $\sum_{l \in \mathcal{N}_k} c_t(l, k) = 1, \quad c_t(l, k) \geq 0, \quad c_t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k,$

where $c_t(l, k)$ is the aggregation weight assigned by worker k to worker l , computed based on the aggregation rule being utilized. Note that any aggregation method used in P2P learning can be expressed in this form. For example, in case of average aggregation, $c_t(l, k) = \frac{1}{|\mathcal{N}_k|}$

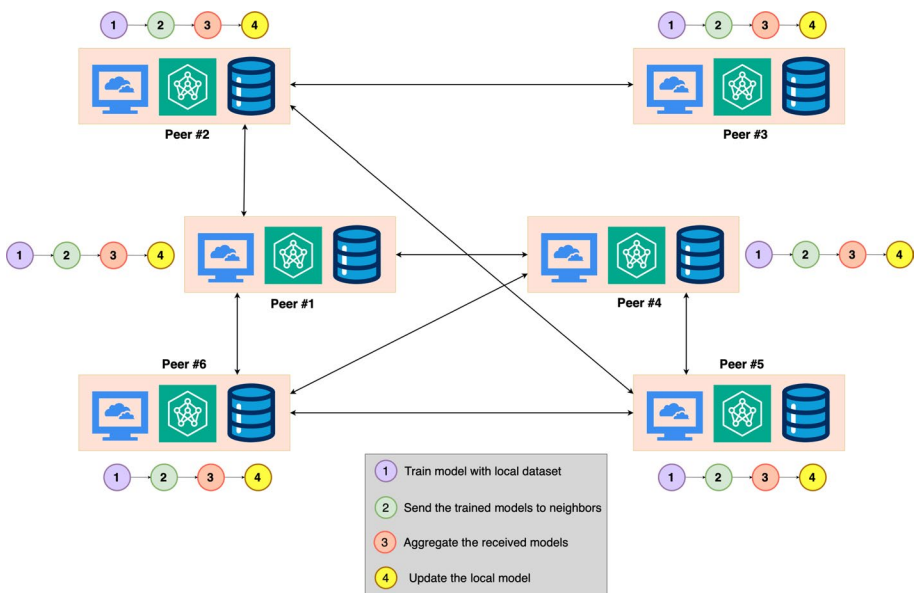


Fig. 1 Schematic of peer-to-peer learning framework

for all $l \in \mathcal{N}_k$. For medoid aggregation, $c_l(l_1, k) = 1$ for an $l_1 \in \mathcal{N}_k$ that has the minimum energy among the neighbors, and $c_l(j, k) = 1$ for all $j \in \mathcal{N}_k \setminus \{l_1\}$. The P2P learning workflow is demonstrated in Fig. 1.

We consider that F out of N workers can be adversarial. The sets of normal workers is denoted as $J' \subset J$. The adversarial peers share malicious parameters with their neighbors instead of true parameters. The malicious values depend on the type of attack. Despite the fact that collaboration between peers is intended to improve learning performance, peers must exercise caution to ensure that the information from malicious neighbors does not affect their learning. The resilient aggregation technique is employed to ensure this, where the aggregation weights are so designed that the learning of normal workers remains unaffected when the number of malicious neighbors is below a specific threshold. The resilience of the aggregation depends on the redundancy of the communication graph. In our theoretical analysis, this requirement is captured by Assumption 5, which imposes a source-component condition on all reduced graphs.

3.2 Attack Model

This paper focuses on model poisoning attacks, where the adversarial workers share malicious information during broadcast phase (Bagdasaryan et al., 2020). We assume that the adversary has the ability to alter the true parameter values of the attacked workers to different values, aimed to compromise the learning performance of the whole learning system. In the following two attack models, the adversary does not need to have knowledge of the aggregation rule used by the peers or the total number of workers in the network.

- Sign-flipping (SF) attack: the parameters shared by the adversarial workers are altered in direction. The parameters shared by the workers are given as follows:

$$\hat{w}_{k,att}^t = \begin{cases} \hat{w}_k^t & \text{if } k \in J' \\ -\sigma_{SF} \cdot \hat{w}_k^t & \text{otherwise} \end{cases} \quad (4)$$

where $\sigma_{SF} \in \mathbb{R}^+$ is the attack strength.

- Arbitrary Byzantine attack: the adversarial workers share arbitrary parameters of matching dimensions. The shared parameters among the neighbors in this case are given as follows:

$$\hat{w}_{k,att}^t = \begin{cases} \hat{w}_k^t & \text{if } k \in J' \\ * & \text{otherwise} \end{cases} \quad (5)$$

3.2.1 Time-Coupled Attacks

These state-of-the-art attacks involve sending small perturbations over multiple epochs, with the goal of destabilizing the learning process. Compared to the attacks discussed above, these sophisticated attacks require the adversary to have additional knowledge regarding the total number of workers, the number of benign workers, and the aggregation rule used by

the workers. For time-coupled attacks, the parameters shared by an adversarial worker are expressed as

$$\hat{w}_{k,att}^t = \bar{w}_k^t + \sigma_{TC} \cdot a^t, \quad (6)$$

where $\bar{w}_k^t$ are the estimated true parameters, given by the average of the normal workers' parameters, σ_{TC} is the attack strength and a^t is the attack direction vector, determined by the specific attack type.

- **Fall-of-empires (FoE) or inner-product manipulation (IPM) attack:** this modifies the worker parameters in such a way that the inner product between the true parameters and the aggregated parameters becomes negative. For this attack, the direction vector is chosen as $a^t = -\bar{w}_k^t$, thus

$$\begin{aligned} \hat{w}_{k,att}^t &= \bar{w}_k^t - \sigma_{TC} \cdot \bar{w}_k^t \\ &= (1 - \sigma_{TC})\bar{w}_k^t \end{aligned} \quad (7)$$

- **A-little-is-enough (ALIE) attack:** this is devised by consistently applying small changes to the worker parameters. This works by finding the perturbation range of the worker parameters for which the adversarial ones are not detected by the aggregation. Then the adversary applies this perturbation to the average of the normal worker parameters to compute $\hat{w}_{k,att}^t$.

After $\hat{w}_{k,att}^t$ is computed, it is broadcast by every adversarial worker.

4 Similarity-Based Resilient Adaptive Aggregation

4.1 Assumptions

The proposed resilient P2P learning algorithm is based on the following assumptions:

Assumption 1 The risk function $r_k(\cdot)$ is non-convex and a.s. twice differentiable in the first argument. Let $\mathcal{W}_s^*$ denote the set of all first-order stationary points of the statistical risk $R_k(w) = \mathbb{E}_\xi [r_k(w)]$, i.e., $\mathcal{W}_s^* = \{w \in \mathbb{R}^d : \nabla R_k(w) = 0\}$. Then for any $w_s^* \in \mathcal{W}_s^*$, the statistical risk is locally m -strongly convex in a sufficiently large neighborhood of w_s^* , i.e., there exist two positive constants m and β such that $\forall w \in \mathbb{B}(w_s^*, \beta)$, $\nabla^2 R_k(w) \geq mI$, where $\nabla^2 R_k(w)$ is the Hessian of $R_k(w)$.

Assumption 2 The risk function $r_k(\cdot)$ has $\mathcal{L}$ -Lipschitz continuous gradient; i.e., $\forall w_1, w_2$, $\|\nabla r(w_1; (x, y)) - \nabla r(w_2; (x, y))\| \leq L \|w_1 - w_2\|$, where $L > 0$ is the Lipschitz constant.

Assumption 3 For every normal worker $k \in \mathcal{V}_c$, the stochastic gradient $\nabla \ell(w_k^t; \xi_k^t)$ is an unbiased estimate of $\nabla r_k(w_k^t)$, i.e., $\mathbb{E} [\nabla \ell(w_k^t; \xi_k^t)] = \nabla r_k(w_k^t)$, for all $t \in \mathbb{N}$.

Assumption 4 For every normal worker $k \in \mathcal{V}_c$, there exists $a_k \geq 1$, such that for all $t \in \mathbb{N}$,

$$\mathbb{E} \left[\left\| \nabla \ell(w_k^t; \xi_k^t) \right\|_2^2 \right] \leq M + a_k \left\| \nabla r_k(w_k^t) \right\|_2^2.$$

Assumption 5 All reduced graphs $\mathcal{G}_r$ generated from $\mathcal{G}(J, \mathcal{E})$ contain a source component of cardinality at least $(F + 1)$.

Assumption 1 guarantees local strong convexity of the risk function, but it does not imply global convexity. This assumption is widely used in the literature for analysis of non-convex optimization in ML (Jain & Kar, 2017). It helps in proving local convergence in the sense that they hold as long as the parameters of the workers are initialized within a sufficiently small neighborhood of a stationary point. Assumption 2 implies that the risk function itself is also $\mathcal{L}$ -Lipschitz continuous, i.e., $\forall w_1, w_2, \|r(w_1; (x, y)) - r(w_2; (x, y))\| \leq L' \|w_1 - w_2\|$, where $L' > 0$ is the Lipschitz constant. Assumption 5 is used to ensure that there is sufficient graph redundancy for information from normal workers to propagate despite Byzantine workers. An efficient certification of this assumption remains an open problem for arbitrary communication graphs. Rather, this assumption should be interpreted as a graph-robustness condition required by the analysis. Erdős–Rényi (ER) graphs (used for evaluation in this paper) typically satisfy this type of reduced-graph connectivity condition when the ratio between the average incoming degree and the number of Byzantine nodes is sufficiently large (Yang & Bajwa, 2019). Therefore, in this work, Assumption 5 specifies the class of communication graphs for which the theoretical guarantees apply.

4.2 Adaptive Aggregation Rule

The proposed approach aims to learn the optimal aggregation weights for each normal worker k , by minimizing the quadratic distance between the aggregated model parameters w_k^t and the optimal parameters w_k^* , i.e., $\min_{C_k^t} \|w_k^t - w_k^*\|^2$, where $C_k^t = \{c_t(1, k), c_t(2, k), \dots, c_t(N, k)\}^\top$ is the vector with aggregation weights assigned by worker k to all other workers. Using (3), the optimization problem is formulated as

$$\begin{aligned} & \min_{C_k^t} \left\| \sum_{l \in \mathcal{N}_k} (c_t(l, k) \hat{w}_l^t) - w_k^* \right\|^2 \\ & \text{subject to } \sum_{l \in \mathcal{N}_k} c_t(l, k) = 1, \quad c_t(l, k) \geq 0, \quad c_t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k. \end{aligned} \quad (8)$$

The optimal aggregation weights are determined by solving (8), the derivation is presented below.

4.2.1 Solving the Optimization Problem

Using $\sum_{l \in \mathcal{N}_k} c_t(l, k) = 1$, we can manipulate the optimization function as

$$\begin{aligned}\sum_{l \in \mathcal{N}_k} (c_t(l, k) \hat{w}_l^t) - w_k^* &= \sum_{l \in \mathcal{N}_k} (c_t(l, k) \hat{w}_l^t) - \sum_{l \in \mathcal{N}_k} (c_t(l, k) w_k^*) \\ &= \sum_{l \in \mathcal{N}_k} c_t(l, k) (\hat{w}_l^t - w_k^*)\end{aligned}$$

After taking norm, we may write

$$\left\| \sum_{l \in \mathcal{N}_k} c_t(l, k) (\hat{w}_l^t - w_k^*) \right\| \leq \sum_{l \in \mathcal{N}_k} c_t(l, k) \left\| \hat{w}_l^t - w_k^* \right\|$$

as $c_t(l, k) \geq 0$. Using Cauchy-Schwarz inequality, we may write

$$\left(\sum_{l \in \mathcal{N}_k} c_t(l, k) \left\| \hat{w}_l^t - w_k^* \right\| \right)^2 \leq |\mathcal{N}_k| \sum_{l \in \mathcal{N}_k} c_t(l, k)^2 \left\| \hat{w}_l^t - w_k^* \right\|^2$$

Using the above discussion, we can transform the optimization function to $|\mathcal{N}_k| \sum_{l \in \mathcal{N}_k} c_t(l, k)^2 \left\| \hat{w}_l^t - w_k^* \right\|^2$. Equivalently, the optimization problem becomes

$$\begin{aligned}\min_{C_k^t} \quad & \sum_{l \in \mathcal{N}_k} c_t(l, k)^2 \left\| \hat{w}_l^t - w_k^* \right\|^2 \\ \text{subject to} \quad & \sum_{l \in \mathcal{N}_k} c_t(l, k) = 1, \quad c_t(l, k) \geq 0, \quad c_t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k.\end{aligned}\tag{9}$$

We assume that there exists a positive constant Γ such that for all $j \in J'$ and $t \in \mathbb{N}$, $\left\| w_j^t - w_s^* \right\| \leq \Gamma$, where $w_s^* \in \mathcal{W}_s^*$. Assuming that w_j^0 for all $j \in J'$ is initialized within $\mathbb{B}(w_s^*, \Gamma)$, the risk function of each worker acts as m -strictly convex, as per Assumption 1. Thus, it holds that

$$r_k(\hat{w}_l^t) - r_k(w_k^*) \geq \langle \nabla r_k(w_k^*), y - x \rangle + \frac{m}{2} \left\| \hat{w}_l^t - w_k^* \right\|^2.$$

Since $\nabla r_k(w_k^*) = 0$, we obtain

$$\left\| \hat{w}_l^t - w_k^* \right\|^2 \leq \frac{2}{m} (r_k(\hat{w}_l^t) - r_k(w_k^*)).\tag{10}$$

Instead of minimizing $\left\| \hat{w}_l^t - w_k^* \right\|^2$, we aim to minimize its upper bound, as given by (10). Thus, we reformulate the optimization problem (8) as¹

¹The factor of $2/m$ is removed for brevity of the solution, without affecting the outcome of the optimization.

$$\begin{aligned} & \min_{C_k^t} \sum_{l \in \mathcal{N}_k} c_t(l, k)^2 \cdot (r_k(\hat{w}_l^t) - r_k(w_k^*)) \\ & \text{subject to } \sum_{l \in \mathcal{N}_k} c_t(l, k) = 1, \quad c_t(l, k) \geq 0, \quad c_t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k. \end{aligned} \quad (11)$$

As $r_k(w_k^*)$ is small compared to $r_k(\hat{w}_l^t)$, we use $r_k(w_k^*) = 0$, which gives us the modified optimization problem²

$$\begin{aligned} & \min_{C_k^t} \sum_{l \in \mathcal{N}_k} \frac{1}{2} c_t(l, k)^2 \cdot r_k(\hat{w}_l^t) \\ & \text{subject to } \sum_{l \in \mathcal{N}_k} c_t(l, k) = 1, \quad c_t(l, k) \geq 0, \quad c_t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k. \end{aligned} \quad (12)$$

After incorporating the constraint $\sum_{l \in \mathcal{N}_k} c_t(l, k) = 1$, the Lagrangian of (12) is obtained as

$$\mathcal{L}(c_t(l, k), \lambda) = \sum_{l \in \mathcal{N}_k} \frac{1}{2} c_t^2(l, k) r_k(\hat{w}_l^t) + \lambda \left(1 - \sum_{l \in \mathcal{N}_k} c_t(l, k) \right), \quad (13)$$

where λ is the Lagrange multiplier. Now, taking the gradient of the Lagrangian w.r.t. $c_t(l, k)$ and equating it to zero, we get

$$c_t(l, k) r_k(\hat{w}_l^t) - \lambda = 0, \quad \forall l \in \mathcal{N}_k. \quad (14)$$

This yields

$$c_t(l, k) = \frac{\lambda}{r_k(\hat{w}_l^t)}, \quad \forall l \in \mathcal{N}_k.$$

Using this in the constraint, we get

$$\lambda \sum_{l \in \mathcal{N}_k} \frac{1}{r_k(\hat{w}_l^t)} = 1.$$

Thus, the Lagrange multiplier is given by

$$\lambda = \frac{1}{\sum_{l \in \mathcal{N}_k} r_k(\hat{w}_l^t)^{-1}}.$$

Substituting this in (14), we get the optimal aggregation weights as

²The factor of 1/2 is introduced for simplification of solution.

$$c_t(l, k) = \begin{cases} \frac{r_k(\hat{w}_l^t)^{-1}}{\sum_{p \in \mathcal{N}_k} r_k(\hat{w}_p^t)^{-1}} & \text{for } l \in \mathcal{N}_k, \\ 0 & \text{for } l \notin \mathcal{N}_k. \end{cases} \quad (15)$$

Remark 1 The derived optimal aggregation weights in (15) are computed at each worker locally, using its local data and the parameters obtained from the neighbors. This follows the data privacy policy, while judging the similarity between the learning behavior of the neighbors with itself.

4.2.2 Selective Collaboration with Better-Performing Neighbors

The aggregation weights derived from the optimization problem require each agent to aggregate parameters from all of its neighbors, i.e., (15) is obtained when the feasible aggregation set is the full neighborhood $\mathcal{N}_k$. However, since the goal here is to promote similarity toward better-performing models, it is beneficial for an agent to selectively interact with neighbors that exhibit lower risk. To achieve this, each agent aggregates only the parameters of those neighbors whose models yield a smaller risk than its own, while retaining the original form of the aggregation weights. The selective-collaboration rule used in the proposed algorithm can be interpreted as solving the same optimization problem under an additional feasibility constraint. Specifically, worker k restricts the feasible aggregation set to

$$\mathcal{N}_k^s(t) = \{l \in \mathcal{N}_k : r_k(\hat{w}_l^t) \leq r_k(\hat{w}_k^t)\}.$$

The corresponding restricted optimization problem is

$$\min_{C_k^t} \left\| \sum_{l \in \mathcal{N}_k^s(t)} c^t(l, k) \hat{w}_l^t - w_k^* \right\|^2$$

subject to

$$\sum_{l \in \mathcal{N}_k^s(t)} c^t(l, k) = 1, \quad c^t(l, k) \geq 0, \quad c^t(l, k) = 0 \text{ for } l \notin \mathcal{N}_k^s(t).$$

Following the same upper-bound argument used in (9)–(12), this becomes

$$\min_{C_k^t} \sum_{l \in \mathcal{N}_k^s(t)} \frac{1}{2} c^t(l, k)^2 r_k(\hat{w}_l^t),$$

subject to the constraints over $\mathcal{N}_k^s$. Solving thus constrained problem using similar flow of derivation as used earlier, the updated aggregation weights are found as:

$$c_t(l, k) = \begin{cases} \frac{r_k(\hat{w}_l^t)^{-1}}{\sum_{p \in \mathcal{N}_k^s} r_k(\hat{w}_p^t)^{-1}} & \text{for } l \in \mathcal{N}_k^s, \\ 0 & \text{otherwise,} \end{cases} \quad (16)$$

Thus, (16) is the optimizer of the restricted version of the aggregation objective, where the restriction implements selective collaboration with neighbors that do not increase worker k 's local empirical risk. By restricting aggregation to better-performing neighbors, this strategy encourages each agent to progressively align its model toward lower-risk solutions while mitigating the influence of poorly performing or potentially malicious peers.

Remark 2 The derived aggregation rule uses the local m -strong convexity of the risk function of the workers. As mentioned earlier, this does not mean that the risk function is globally convex. Local convexity is widely used in literature for analysis in case of non-convex scenarios.

Remark 3 The aggregation rule in (16) is derived without imposing any assumption on the data distribution across workers. Therefore, the proposed algorithm can be applied to both iid and non-iid data distributions. Although the proposed method is primarily motivated by decentralized learning under non-iid data distributions, and is the main focus of this work, it is also applicable to iid settings. In the iid case, the local data distributions across normal workers are more homogeneous, which generally reduces the statistical heterogeneity faced by decentralized learning algorithms. As shown in Section 6.2, AdapAgg remains effective under iid data distributions on both MNIST and Spambase, maintaining strong test accuracy under multiple attacks.

4.3 Algorithm and Complexity

The P2P learning algorithm with the proposed aggregation method is outlined in Algorithm 1.

-
- 1: **Input:** number of epochs T , communication graph $\mathcal{G} = \{J, \mathcal{E}\}$, step sizes $\{\rho^t\}_{t \geq 0}$, and initial parameters $\{w_i^0\}$ for all $i \in [1, 2, \dots, N]$.
 - 2: **for** epoch $t = 1$ to T **do**
 - 3: **for** each normal agent $i \in J$ **do**
 - 4: Update $\hat{w}_i^t$ using (2).
 - 5: Send $\hat{w}_i^t$ to the neighbors $\mathcal{N}_i$.
 - 6: **end for**
 - 7: Adversarial workers $i \in J \setminus J'$ send malicious parameters to their neighbors.
 - 8: **for** each normal agent $i \in J$ **do**
 - 9: Compute risk function $r_i(\hat{w}_l^t)$ for $l \in \mathcal{N}_i$.
 - 10: Determine the set of better performing neighbors $\mathcal{N}_i^s$.
 - 11: Compute aggregation weights over $\mathcal{N}_i^s$ using the restricted optimization solution in (16).
 - 12: Compute updated parameters w_i^t using (3).
 - 13: **end for**
 - 14: **end for**
-

Algorithm 1 Peer-to-peer distributed learning with adaptive model aggregation

The time complexity of the adaptive aggregation step can be analyzed as follows. The adaptive aggregation step at worker k consists of three operations: (i) evaluating the local empirical risk of each received neighbor model, (ii) computing the inverse-risk aggregation weights, and (iii) forming the weighted average of the selected neighbor parameters. Let B_s denote the batch size used for scoring neighbors, $C_{\text{fwd}}(B_s)$ denote the cost of one forward-pass loss evaluation on a batch of size B_s , d denote the model dimension, and $|\mathcal{N}_k|$ denote the number of neighbors of worker k . The cost of evaluating the neighbor losses is $O(C_{\text{fwd}}(B_s)|\mathcal{N}_k|)$. Computing the aggregation weights and forming the weighted average costs $O(d \cdot |\mathcal{N}_k|)$. Therefore, the total aggregation-time complexity per worker per round is

$$O(C_{\text{fwd}}(B_s)|\mathcal{N}_k|) + O(d \cdot |\mathcal{N}_k|).$$

The additional cost of the proposed method compared with simple averaging and several coordinate-wise robust aggregation rules comes from the forward-pass evaluations used to score neighbor models. Importantly, these evaluations are performed only on a local scoring batch and require no backpropagation. Therefore, the additional computation is smaller than performing extra local SGD steps. For small neural networks, the $O(d \cdot |\mathcal{N}_k|)$ weighted-averaging term may be comparable to the scoring cost; for larger CNN or ResNet models, the forward-pass term dominates.

The communication overhead of the proposed method is the same order as standard model-aggregation-based P2P learning methods. Each worker broadcasts only its intermediate model parameters to its neighbors, resulting in $O(d \cdot |\mathcal{N}_k|)$ communication per worker per round. The method does not require sharing raw data, labels, gradients, loss values, or additional certificates. Thus, the resilience improvement is obtained through additional local computation rather than additional communication.

In practice, the first term dominates for complex models such as CNN, whereas for small models the overall complexity reduces to $O(d \cdot |\mathcal{N}_k|)$. Compared to existing robust aggregation methods such as coordinate-wise median or geometric median, the proposed adaptive aggregation incurs additional computational overhead due to the forward loss evaluations. This additional cost reflects a fundamental trade-off between computational complexity and resilience guarantees, and is consistent with the stronger resilience properties achieved by the proposed method compared to the existing rules. Table 1 compares the aggregation-stage overhead of the proposed method with the baseline aggregation rules. The proposed method has the same communication order as the model-aggregation baselines, but incurs additional local computation due to the loss evaluations used for similarity-aware weighting.

Table 1 Communication and computation comparison between aggregation rules

Aggregation rule	Local aggregation computation	Communication per round
Vanilla averaging	$O(d \mathcal{N}_k)$	$O(d \mathcal{N}_k)$
Coordinate-wise median	$O(d \mathcal{N}_k \log \mathcal{N}_k)$	$O(d \mathcal{N}_k)$
Trimmed mean	$O(d \mathcal{N}_k \log \mathcal{N}_k)$	$O(d \mathcal{N}_k)$
Krum	$O(d \mathcal{N}_k ^2)$	$O(d \mathcal{N}_k)$
Centered clipping	$O(I_{\text{clip}}d \mathcal{N}_k)$	$O(d \mathcal{N}_k)$
Nearest-neighbor averaging	$O(d \mathcal{N}_k \log \mathcal{N}_k)$	$O(d \mathcal{N}_k)$
AdapAgg	$O(C_{\text{fwd}}(B_s) \mathcal{N}_k) + O(d \mathcal{N}_k)$	$O(d \mathcal{N}_k)$

5 Theoretical Analysis

In this section, first we prove that the normal workers' parameters asymptotically reach a consensus, converging to the same value, despite decentralized operation, non-i.i.d. data, nonconvex loss and adversarial neighbors. Next, we establish optimal guarantees with respect to the local optimal parameters of each normal worker. This involves proving that the model parameters of normal workers converge to a neighborhood of their respective local optima with a bounded error. Both of these proofs hold under Assumption 5.

5.1 Consensus of Normal Nodes

This analysis follows the notations J and J' as defined earlier. Additionally, the sets of normal and adversarial neighbors of worker j are denoted by $\mathcal{N}_j^s$ and $\mathcal{N}_j^b$, respectively, such that $\mathcal{N}_j = \mathcal{N}_j^s \cup \mathcal{N}_j^b$. While $\mathcal{N}_j^s$ was defined earlier to represent the neighbors with smaller risk than worker j , we now introduce $\mathcal{N}_j^l$ to denote the set of neighbors with larger risk, evaluated on the local data; that is, for $i \in \mathcal{N}_j^l$, $r_j(\hat{w}_i^t) > r_j(\hat{w}_j^t)$. Note that the sets $\mathcal{N}_j^s$, $\mathcal{N}_j^b$, $\mathcal{N}_j^s$, and $\mathcal{N}_j^l$ are time-varying, but for notational simplicity, the dependency on t is omitted.

To prove consensus of the normal workers' parameters, we first define a vector $\Omega(t) \in \mathbb{R}^{J' \times P}$ by stacking the parameters of the normal workers, i.e., $[\Omega(t)]_j = w_j^t$. Similarly, define $\hat{\Omega}(t)$ such that $[\hat{\Omega}(t)]_j = \hat{w}_j^t$. Also, define the set $G(t) \in \mathbb{R}^{J' \times P}$ such that $[G(t)]_j = \nabla l(w_j^{t-1}; \xi_j^{t-1})$.

The first step of the consensus analysis is to represent the parameter update of the normal workers in terms of other normal workers' updates, which is given in the following lemma.

Lemma 1³ *The parameter update at the normal workers can be expressed in the following form:*

$$\Omega(t+1) = Y(t)\Omega(t) - \rho^t \cdot Y(t)G(t), \quad (17)$$

where $Y(t) \in \mathbb{R}^{J' \times |J'|}$ is a matrix whose elements are given as

$$[Y(t)]_{ij} = \begin{cases} c_t(i, j), & \text{if } i \in \mathcal{N}_j^s \cap \mathcal{N}_j^s \\ \sum_{i \in \mathcal{N}_j^s \cap \mathcal{N}_j^b} \theta_j^t c_t(i, j), & \text{if } i \in \mathcal{N}_j^l \cap \mathcal{N}_j^s \\ \sum_{i \in \mathcal{N}_j^s \cap \mathcal{N}_j^b} (1 - \theta_j^t) c_t(i, j), & \text{if } i \in \mathcal{N}_j \setminus \{j\} \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

and $\theta_j^t \in (0, 1)$ is a scalar.

Lemma 1 demonstrates that the parameter aggregation of every normal worker can be expressed as a combination of parameters of other normal workers. Note that the consensus proof does not require the knowledge of $Y(t)$, moreover, it cannot be guaranteed that $Y(t)$ is unique. However, as evident from its definition, $Y(t)$ is a row-stochastic matrix, which will be used in the remaining part of this analysis.

³ All proofs are included in the appendix.

In an attempt to express (17) in the form of a state-transition equation, we fix an arbitrary $t_0 \geq 0$ and consider $t \geq t_0$. It then follows from (17) that

$$\begin{aligned}\Omega(t+1) &= Y(t)\Omega(t) - \rho^t \cdot Y(t)G(t) \\ &= Y(t)Y(t-1) \cdots Y(t_0)\Omega(t_0) - \rho^t \cdot Y(t)G(t) \\ &\quad - \sum_{\tau=t_0}^{t-1} Y(t)Y(t-1) \cdots Y(\tau+1)\rho^\tau \cdot Y(\tau)G(\tau).\end{aligned}\quad (19)$$

Define

$$\begin{aligned}\Phi(t, t_0) &= Y(t)Y(t-1) \cdots Y(t_0) \\ \Phi(t, t) &= Y(t) \\ \Phi(t, t+1) &= \mathbb{I}.\end{aligned}$$

$\Phi(t, t_0)$, being a product of row-stochastic matrices, is a row stochastic matrix itself. Then (19) is expressed as

$$\Omega(t+1) = \Phi(t, t_0)\Omega(t_0) - \sum_{\tau=t_0}^t \Phi(t, \tau+1)\rho^\tau \cdot Y(\tau)G(\tau) \quad (20)$$

Now, we state two key properties of $\Phi(t, t_0)$ from Su and Vaidya (2015) that are required for the consensus analysis. These properties are states as lemmas as below.

Lemma 2 *Su and Vaidya (2015) Under Assumption 5, for any $t_0 \geq 0$, there exists a stochastic vector $\Pi(t_0)$ such that*

$$\lim_{t \rightarrow \infty} \Phi(t, t_0) = \mathbf{1} \cdot \Pi^\top(t_0) \quad (21)$$

Lemma 2 states that the product of row-stochastic matrices $Y(t)$ converges to a steady-state matrix with identical and stochastic rows.

To characterize the rate of the convergence, let ψ denote the number of reduced graphs generated from the original graph $\mathcal{G}$, and define $\nu := \psi|J'|$ and $\mathcal{N}_{\max} = \max_{j \in J'} |\mathcal{N}_j|$, and

$$\psi := 1 - \frac{1}{(2\mathcal{N}_{\max} - 2b + 1)^\nu}.$$

Lemma 3 *Su and Vaidya (2015) Under Assumption 5, we have, for all $t_0 \geq 0$,*

$$|[\Phi(t, t_0)]_{ji} - [\Pi(t_0)]_i| \leq \mu^{\left(\frac{t-t_0+1}{\nu}\right)} \quad (22)$$

Lemma 3 describes the rate at which the rows of $\Phi(t, t_0)$ converges to $\Pi(t_0)$.

The lemmas are now used to prove consensus of the normal workers using the adaptive aggregation method. Assuming $t_0 = 1$, we have from (20),

$$\Omega(t+1) = \Phi(t, 1)\Omega(1) - \sum_{\tau=1}^t \Phi(t, \tau+1)\rho^\tau \cdot Y(\tau)G(\tau). \quad (23)$$

$Y(t)G(t)$ is a $|J'| \times P$ matrix of weighted gradients of all normal workers. Suppose normal workers stop computing local gradients at time step t , and use $G(t+t') = 0$ for $t' \geq 0$. Define

$$V(t) := \lim_{t' \rightarrow \infty} \Omega(t+t'+1).$$

Using Lemma 2, we can now manipulate $V(t)$ as

$$\begin{aligned} V(t) &= \lim_{t' \rightarrow \infty} \Phi(t+t', 1)\Omega(1) - \lim_{t' \rightarrow \infty} \sum_{\tau=1}^{t+t'} \Phi(t+t', \tau)\rho^\tau \cdot Y(\tau)G(\tau) \\ &= 1 \cdot \Pi^\top(1) \cdot \Omega(1) - \sum_{\tau=1}^{t-1} 1 \cdot \Pi^\top(\tau)\rho^\tau Y(\tau)G(\tau). \end{aligned} \quad (24)$$

$\Pi^\top(t)Y(t)G(t)$ is the weighted combination of gradient vectors of all normal workers, and multiplying this by 1 replicates this across all J' rows. The consensus of parameters of the normal nodes are given in the following theorem.

Theorem 1 *Let Assumptions 2 and 5 hold. Then for all $j \in J'$, we have $w_j^t \rightarrow V(t)$ as $t \rightarrow \infty$.*

Theorem 1 show that all the normal workers' parameters reach a common value after a certain number of training epochs. Note that this analysis does not use an upper bound on the number of adversarial workers. Thus, consensus is achieved in the presence of an arbitrary number of adversarial workers, provided that the communication graph satisfies Assumption 5. The analysis does not impose an additional algorithm-specific upper bound on the number of adversarial workers; instead, the admissible adversarial configuration is determined by the reduced-graph source-component condition. Additionally, consensus is guaranteed under conditions of non-iid data distribution and non-convex loss function.

In the following part, we prove that, in addition to agreeing on a common value, the normal workers' parameters converge close to their optimal values, in the presence of an arbitrary number of adversarial neighbors.

5.2 Convergence to Optimality

This analysis is based on Assumption 2 that ensures the local strong convexity of the risk function. We show that the aggregated parameters of each normal worker $k \in \mathcal{V}_c$ converge toward optimality, maintaining a bounded gap.

Lemma 4 *Suppose every worker be initialized with parameters within $\mathbb{B}(w_s^*, \Gamma)$. For every normal worker $k \in \mathcal{V}_c$ that uses aggregation step (3) with aggregation weights given by (16), the following inequality holds*

$$\mathbb{E} [r_k(w_k^t) - r_k(w_k^*)] \leq \frac{1}{|\mathcal{N}_k|} \sum_{l \in \mathcal{N}_k} \mathbb{E} [r_k(\hat{w}_l^t) - r_k(w_k^*)] \quad (25)$$

Theorem 2 Suppose Assumptions 1-4 hold and the parameters of each worker be initialized within $\mathbb{B}(w_k^*, \Gamma)$. Then for every normal worker $k \in \mathcal{V}_c$ that runs the local learning step (2), followed by an aggregation step (3) with weights computed by (16), their parameters converge towards optimality w_k^* as

$$\lim_{t \rightarrow \infty} \mathbb{E} [r_k(w_k^t) - r_k(w_k^*)] \leq \frac{\bar{\rho} L \sigma_k^2}{2m}, \quad (26)$$

for fixed stepsize $\bar{\rho} \in \left(0, \frac{1}{La_k}\right]$, in presence of an arbitrary number of adversarial neighbors.

The optimality gap is bounded and is a function of the learning rate, nature of the loss function and the data distribution among workers. The P2P algorithm where workers use a decreasing learning rate also achieves similar results, with the optimality gap being slightly different from what is given in (26).

6 Empirical Evaluation

6.1 Experimental Setup

The evaluation is performed on four classification tasks, under conditions of non-iid data distribution and non-convex loss function.

6.1.1 Tasks and Datasets

The ML tasks and datasets are discussed below.

- **Human Activity Recognition on UCI-HAR Dataset:** Human activity recognition is a popular ML task where the goal is to predict activities of a human based on the sensor readings. We use the UCI HAR dataset⁴ which contains data collected from the accelerometer and gyroscope sensors embedded in smartphones worn by participants during various physical activities. The data is collected from 30 humans performing the following six activities: walking, walking-upstairs, walking-downstairs, sitting, standing, lying-down.

Table 2 Network architecture for activity recognition task

#	Layer(type)	Details
1	Linear	Number of neurons: 100
2	ReLU	—
3	Linear	Number of neurons: 6
4	Softmax	—

⁴<https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones>.

Table 3 Network architecture for digital classification task

#	Layer(type)	Details
1	Conv2d	output channels: 32, kernel size:3, stride:1
2	ReLU	–
3	MaxPool2d	kernel size:2
4	Conv2d	output channels: 64, kernel size:3, stride:1
5	ReLU	–
6	MaxPool2d	kernel size:2
7	Conv2d	output channels: 64, kernel size:3, stride:1
8	ReLU	–
9	MaxPool2d	kernel size:2
10	Linear	number of neurons: 128
11	ReLU	–
12	Linear	number of neurons: 10

Table 4 Network architecture for spam filtering task

#	Layer(type)	Details
1	Linear	number of neurons: 20
2	ReLU	–
3	Linear	number of neurons: 2
4	Softmax	–

We consider a network of 30 worker machines, each representing an individual user. Due to the inherent variability in human behavior, the resulting data distribution across workers is *naturally non-i.i.d.* The private data of each worker is split into 75% training data and 25% testing data. The sensor readings are embedded in the form of a feature vector of length 561, which is the input to a fully connected neural network model. The details of this model are given in Table. 2. For this task, we used cross-entropy loss, a learning rate of 0.01 and a batch-size of 10.

- Digit Classification on MNIST Dataset:** The MNIST dataset⁵ consists of a large collection of 28×28 pixel grayscale images of handwritten digits (0 through 9), along with their corresponding labels. The dataset contains a total of 60000 images, among which 50000 images are used for training, and the remaining 10000 are used for testing. We consider a group of 20 workers for this task. To simulate the heterogeneous data distribution, we use *pathological non-iid* data distribution among them, where only two out of the ten labels are available at each worker (Li et al., 2022). To achieve this, the data is first sorted by digit label, and then divided into 200 shards of size 300. We then assign each of the 20 workers 10 shards. A 5-layer neural network with three convolution layers and two linear connected layers is used for this task. The detailed description of this model is given in Table. 3. Cross-entropy loss is used for training the network. We used Adam optimizer, batch size of 64 and a learning rate of 0.02.
- Spam Filtering:** This is a binary classification problem, where the objective is to determine whether an email message is spam or not. The Spambase dataset⁶ contains a

⁵<https://www.kaggle.com/datasets/hojjatk/mnist-dataset>.

⁶<https://archive.ics.uci.edu/ml/datasets/spambase>.

total of 4601 instances, with each instance representing a single email message, with an appropriate label of 0 (not spam) or 1 (spam). Each instance is described by a feature vector of length 57, which contains numerical attributes derived from the content of the messages.

The data is divided among 10 workers following a distribution-based label-skew. In this, all local datasets contain data samples belonging to both classes, but their distribution varies from one to another (Li et al., 2024). This is called a *practical non-iid* data distribution scenario. We use a connected neural network as the model which is described in Table. 4. Batch-size used here is 20, and the learning rate is 0.01. The network is trained using cross-entropy loss.

- **Object Classification on CIFAR-10 Dataset:** The CIFAR-10 dataset⁷ is a standard benchmark for image classification consisting of 32×32 color images from 10 object categories. The dataset contains 50000 training images and 10000 testing images. We consider a group of 50 workers for this task. To simulate heterogeneous data distribution across these workers, we use a *practical non-iid* data distribution following a Dirichlet label-skew, where each worker receives a different class distribution controlled by the Dirichlet concentration parameter $\alpha > 0$. For the experiments, we set $\alpha = 0.5$, where smaller α corresponds to stronger heterogeneity. For CIFAR-10 classification, we use a ResNet-20 architecture following the standard CIFAR-style design. The network consists of an initial 3×3 convolution layer with 16 channels, followed by three residual stages with 3 residual blocks per stage. The number of channels is doubled at each stage (16, 32, and 64), with spatial downsampling performed at the beginning of the second and third stages. A global average pooling layer is applied before a final fully connected layer with softmax activation. The description of this model is given in Table 5. The evaluation on this dataset with ResNet-20 helps us understand the applicability of the proposed method to a more complex model and a larger dataset.

Table 5 Network architecture for object classification task

#	Layer (type)	Details
1	Conv2d	output channels: 16, kernel size: 3, stride: 1
2	BatchNorm2d	–
3	ReLU	–
4	Residual Block ×3	$2 \times (\text{Conv2d: 16 ch, } 3 \times 3, \text{ stride 1} + \text{BatchNorm2d} + \text{ReLU})$
5	Residual Block ×3	$2 \times (\text{Conv2d: 32 ch, } 3 \times 3, \text{ stride 2/1} + \text{BatchNorm2d} + \text{ReLU})$
6	Residual Block ×3	$2 \times (\text{Conv2d: 64 ch, } 3 \times 3, \text{ stride 2/1} + \text{BatchNorm2d} + \text{ReLU})$
7	AvgPool2d	output size: 1×1
8	Linear	number of neurons: 10

⁷<https://www.cs.toronto.edu/kriz/cifar.html>.

Table 6 Human activity recognition task on UCI-HAR dataset when 15 out of 30 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.1672	77.36	59.34	0.1792	18.88	14.08
BRIDGE-M	0.0805	72.95	52.22	0.1792	18.58	10.78
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.1676	76.42	50.60	0.1792	19.37	15.49
MoNNA	—	—	—	0.1792	17.50	13.48
AdapAgg (ours)	0.0912	82.83	58.16	0.0126	96.59	82.80
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.1792	18.69	11.24	0.1792	18.02	10.78
BRIDGE-M	0.1792	15.79	7.45	0.1792	16.66	10.00
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.1792	18.98	12.00	0.1792	19.30	10.98
MoNNA	—	—	—	—	—	—
AdapAgg (ours)	0.0191	94.53	77.45	0.0066	96.85	92.23

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as a random graph with edge probability 0.5. The table reports the average training loss, average test accuracy, and minimum test accuracy across normal workers

6.1.2 Attack Models

Four general classes of model poisoning attacks are considered: sign-flipping (SF), arbitrary byzantine, fall-of-empires (FoE) and a-little-is-enough (ALIE). For SF attack, the attack strength σ is chosen as 1. In case of arbitrary Byzantine attack, random signals from the range $(-0.5, 0.5)$ are multiplied with the actual parameters.

For each task, we simulate three attacked scenarios, where 20%, 40% and 50% of the total number of workers are controlled by the adversary.

6.1.3 Baseline Aggregation

We consider three variants of the BRIDGE algorithm (Fang et al., 2022), BRIDGE-T (trimmed mean aggregation), BRIDGE-M (coordinate-wise mean aggregation) and BRIDGE-T (Krum aggregation). Additionally, we simulate the vanilla-P2P algorithm that uses simple averaging to aggregate neighbors' parameters. The following state-of-the-art aggregation techniques are also used as baselines here: P2P learning with centered clipping (CC)(He et al., 2022) aggregation and MoNNA(Farhadkhani et al., 2023) with nearest neighbor aggregation.

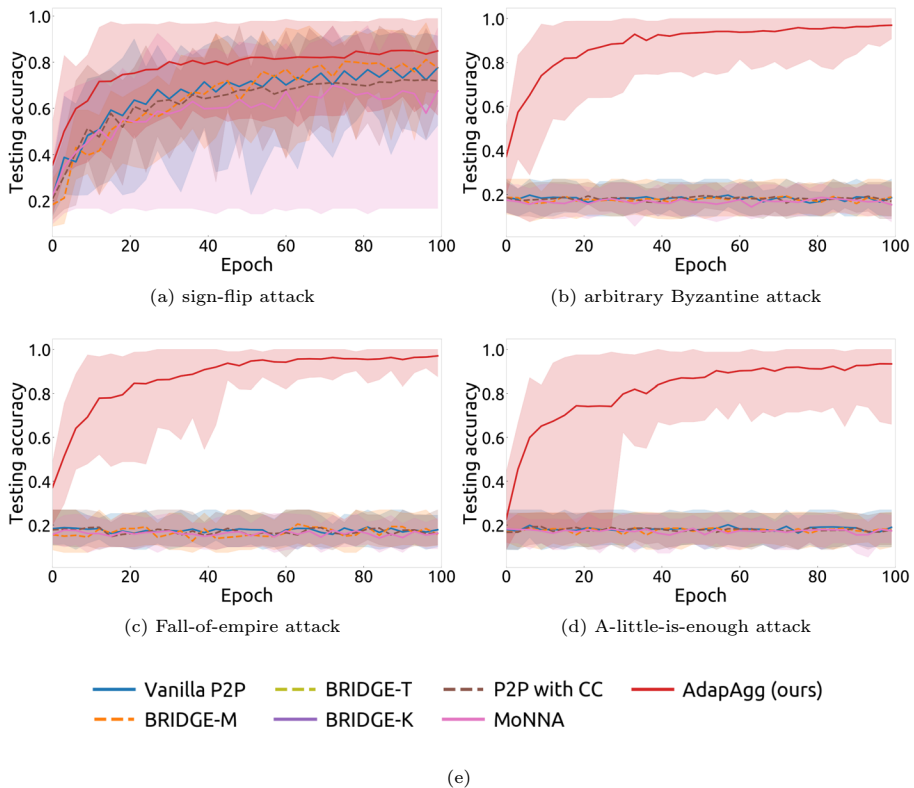


Fig. 2 Test accuracy of activity recognition task in adversarial scenarios under various attacks: 30 total workers, 12 adversarial workers, communication graph: ER with edge probability of 0.5

6.1.4 Worker Network

We use Erdős–Rényi graphs to model the interaction between the workers, where the edges are randomly generated based on a given probability. We generate the communication networks with probability value of 0.5.

For the activity recognition and digit classification tasks, the training is performed for 100 epochs. For spam detection, the training is performed for 200 epochs.

6.2 Results and Discussion

The algorithm is evaluated by reporting the training loss and testing accuracy at the end of the training, the scenarios involve various tasks, attack types, number of adversarial workers and communication graph. For conciseness, we present the results for the most adversarial scenario, where 50% of the workers are compromised. Results for scenarios with 20% and 40% adversarial workers are provided in Appendix B.

Table 7 Digit classification task on MNIST dataset when 10 out of 20 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0360	12.47	0.00	0.0360	12.59	0.00
BRIDGE-M	0.0000	99.82	99.61	0.0000	99.75	99.61
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0360	7.61	0.00	0.0360	12.35	0.00
MoNNA	0.0360	10.03	0.00	0.0360	17.54	0.00
AdapAgg (ours)	0.0000	99.80	99.70	0.0000	99.77	99.50
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0360	20.23	0.00	0.0360	14.80	0.00
BRIDGE-M	0.0000	99.75	99.58	0.0000	99.79	99.57
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0360	17.40	0.00	0.0360	15.23	0.00
MoNNA	0.0360	15.42	0.00	0.0360	13.02	0.00
AdapAgg (ours)	0.0000	99.78	99.56	0.0000	99.67	99.42

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as a random graph with edge probability 0.5

6.2.1 Activity Recognition Task on UCI-HAR

Each worker of the group is evaluated based on the final training loss and testing accuracy. We report the maximum training loss, minimum test accuracy, and average test accuracy across normal workers. The minimum accuracy captures the worst-case normal worker and reflects resilience, while the average accuracy captures the overall performance of the normal-worker population. Table. 6 reports the maximum training loss, minimum test accuracy, and average test accuracy across normal workers for UCI-HAR dataset when 15 out of 30 workers are adversarial.

As evident from the table, our proposed aggregation method ensures more than 74% improved accuracy compared to the baselines in three attack types when half of the workers are adversarial. Note that methods such as trimmed-mean and Krum cannot be used here, as the normal workers do not have enough normal neighbors to be able to apply these methods. Coordinate-wise median, centered clipping and NNA are computed even though their analysis do not support their applicability in such scenarios. The entries with - denote that the algorithm did not converge in those scenarios.

The learning behaviors are better presented by plotting the test accuracy with epochs. Figure 2 plots the test accuracy for the activity recognition task under various attacks when 12 out of 30 workers are adversarial. The average accuracy of the group is plotted as the solid line, while the deviation across the group is depicted by the shaded area.

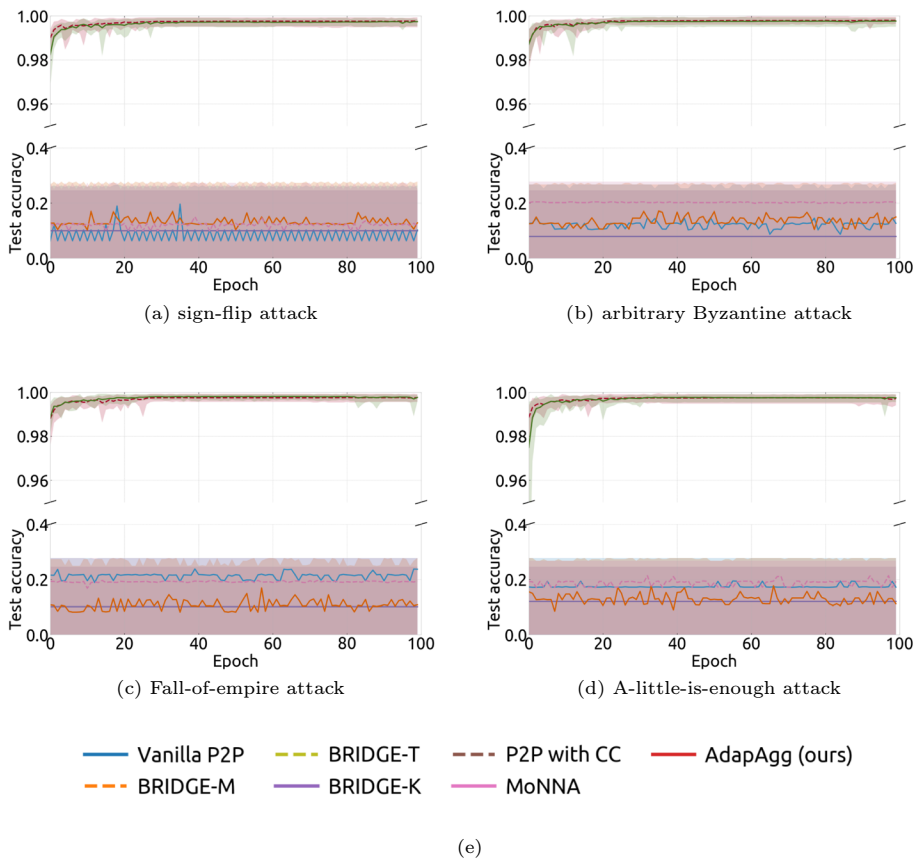


Fig. 3 Test accuracy of digit classification task in adversarial scenarios under various attacks: 20 total workers, 8 adversarial workers, communication graph: ER with edge probability of 0.5

6.2.2 Digit Classification Task on MNIST

The maximum training loss, average test accuracy, and minimum test accuracy for the normal workers are shown in Table 7, when 10 out of 20 workers are under adversarial. For this task, coordinate-wise median performs better than others in most scenarios, while the adaptive method's performance matches it closely. No other aggregation can show moderately well behavior in all attack scenarios.

Similar observation is obtained when 8 out of total 20 workers are adversarial. The test accuracy for the workers under various attack types are plotted in Fig. 3.

6.2.3 Spam Detection Task on Spambase

The maximum training loss, average test accuracy, and minimum test accuracy for spam detection task are presented in Table 8. The adaptive aggregation performs consistently well in case of all attacks, along with CM and NNA.

Table 8 Spam classification task on Spambase dataset when 5 out of 10 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0361	39.35	0.00	0.0356	71.52	51.09
BRIDGE-M	0.0168	97.17	90.22	0.0173	95.00	89.13
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0351	48.91	0.00	0.0358	70.43	50.00
MoNNA	0.0169	93.04	82.61	0.0169	97.61	92.39
AdapAgg (ours)	0.0194	90.00	81.52	0.0170	93.26	82.61
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0357	65.22	1.09	0.0356	69.35	21.74
BRIDGE-M	0.0169	94.13	82.61	0.0169	94.13	82.61
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0352	68.04	50.00	0.0352	65.65	0.00
MoNNA	0.0168	97.17	90.22	0.0169	93.04	82.61
AdapAgg (ours)	0.0176	92.39	82.61	0.0175	95.87	90.22

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as a random graph with edge probability 0.5

The test accuracy of all workers under adversarial conditions, where 4 out of 10 workers are compromised, is presented in Fig. 4. The average accuracy of the 10 workers is plotted here, while the shaded area represents the range of accuracy obtained by other workers. It is evident that the adaptive method outperforms the baseline aggregation methods in all cases.

6.2.4 Object Recognition Task on CIFAR-10

The results for CIFAR-10 dataset are reported in Table 9, when 25 out of 50 workers are adversarial.

As expected, vanilla P2P exhibits noticeable performance degradation under all the attack types. Although BRIDGE-M improves robustness under sign-flip attacks, its performance deteriorates under Byzantine attack. Similarly, P2P with CC and MoNNA offer limited resilience, providing only marginal improvements over the vanilla approach under Byzantine and fall-of-empire attacks. In contrast, the proposed AdapAgg method consistently demonstrates strong robustness across all attack scenarios, achieving the highest test accuracy under both Byzantine and A-Little-Is-Enough attacks. Notably, AdapAgg substantially outperforms all baselines even under the fall-of-empire attack, highlighting the effectiveness of adaptive, similarity-aware aggregation in highly adversarial peer-to-peer learning settings.

6.2.5 Results for a Denser Communication Network

To assess the robustness of the proposed method with respect to the connectivity structure of the worker network, we repeated the experiments using a denser graph with an edge proba-

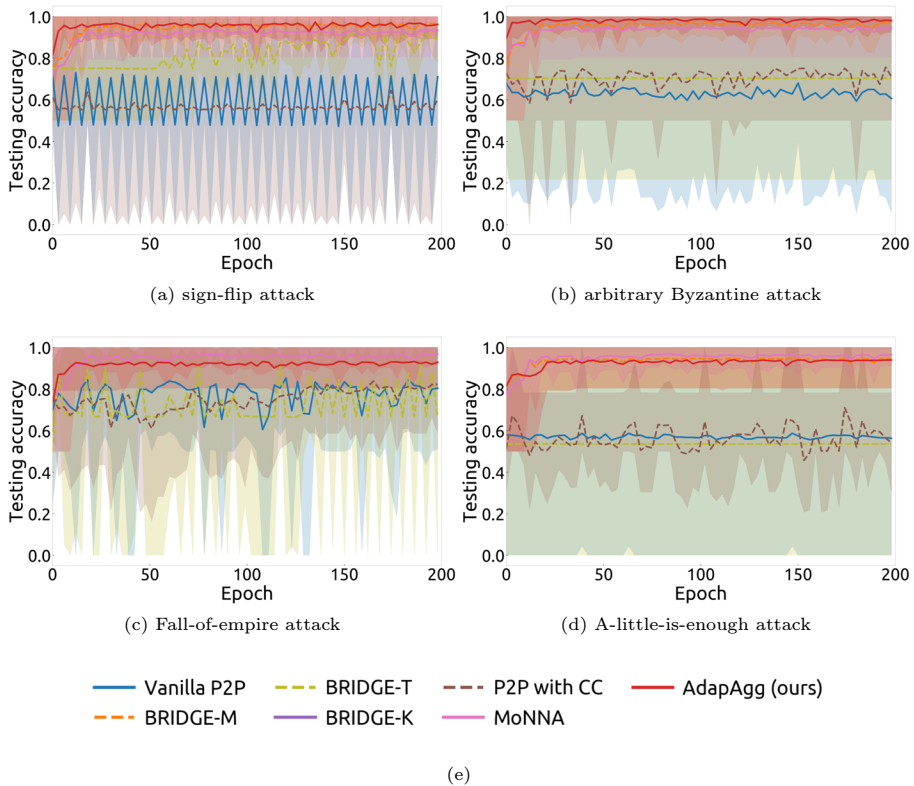


Fig. 4 Test accuracy of spam detection task in adversarial scenarios under various attacks: 10 total workers, 4 adversarial workers, communication graph: ER with edge probability of 0.5

bility of 0.8. This models a scenario where each worker communicates with more neighbors compared to the previous setting. Even under this denser topology, the adaptive aggregation method consistently outperforms baseline approaches, with comparable performance in a few cases. For example, in case of human activity recognition, the adaptive aggregation method shows an improvement of over 74% absolute points over any other aggregation methods, under ALIE attack. Under other types of attack, this shows an improvement by at least 15% compared to other baselines. Detailed results are provided in Appendix B.

6.2.6 Results Under iid Data Distributions

While our main experiments focus on non-iid data distributions, which are more challenging and practically relevant in decentralized learning, we also evaluate the proposed method under iid data distributions to support the claim in Remark 3. In this setting, the local training samples are uniformly distributed across the normal workers. We consider two benchmark datasets, MNIST and Spambase, and use the same attack models and network topology as in the non-iid experiments.

Table 9 Object classification task on CIFAR-10 dataset using ResNet-20 when 25 out of 50 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	1.82	41.6	33.09	2.21	11.3	7.98
BRIDGE-M	1.04	68.2	42.15	2.18	12.1	7.88
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	1.79	43.8	33.12	2.05	13.7	7.98
MoNNA	1.33	55.6	36.98	1.98	21.4	15.09
AdapAgg (ours)	0.92	72.8	66.90	0.68	78.6	75.23
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	2.26	10.9	7.12	2.19	12.4	8.02
BRIDGE-M	0.68	76.7	64.89	2.11	74.2	56.90
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	2.07	13.6	9.65	2.01	15.9	11.23
MoNNA	1.94	24.3	18.44	1.88	29.10	22.77
AdapAgg (ours)	0.75	76.4	70.11	0.61	80.2	72.94

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as a random graph with edge probability 0.5

Table 10 Digit classification task on MNIST dataset under iid data distribution when 8 out of 20 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0364	10.37	9.90	0.0364	11.23	10.60
BRIDGE-M	0.0000	97.79	96.70	0.0000	98.20	96.70
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0364	10.07	9.00	0.0364	11.39	10.60
MoNNA	0.0012	58.83	38.10	0.0364	11.20	10.10
AdapAgg (ours)	0.0000	97.86	96.40	0.0071	90.04	65.00
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0364	11.36	10.60	0.0364	11.47	10.60
BRIDGE-M	0.0000	90.71	96.10	0.0000	98.41	97.00
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0364	10.86	8.50	0.0364	11.10	10.60
MoNNA	0.0364	10.76	8.50	0.0364	11.04	10.10
AdapAgg (ours)	0.0032	93.13	39.60	0.0001	98.43	97.30

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as an Erdős–Rényi (ER) graph with edge probability 0.5

Table 11 Spam classification task on Spambase dataset under iid data distribution when 4 out of 10 workers are under attack

Algorithm	Sign-flip attack			Byzantine attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0358	40.68	35.87	0.0358	61.02	54.35
BRIDGE-M	0.0190	92.55	90.22	0.0195	92.08	86.96
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0196	53.17	40.22	0.0193	44.25	21.30
MoNNA	0.0198	90.53	85.87	0.0201	93.17	88.04
AdapAgg (ours)	0.0195	93.01	85.87	0.0197	92.39	90.22
Algorithm	Fall-of-empire attack			A-little-is-enough attack		
	Avg loss	Avg Acc	Min Acc	Avg loss	Avg Acc	Min Acc
Vanilla P2P	0.0358	60.09	50.00	0.0358	62.27	55.43
BRIDGE-M	0.0185	91.77	85.87	0.0184	94.25	89.13
BRIDGE-T	—	—	—	—	—	—
BRIDGE-K	—	—	—	—	—	—
P2P with CC	0.0194	51.30	39.22	0.0196	63.79	41.13
MoNNA	0.0204	91.61	90.22	0.0256	64.60	55.43
AdapAgg (ours)	0.0197	91.93	88.04	0.0197	92.86	85.87

Bold numbers in the table are to represent the best accuracy achieved across all aggregation methods

The interaction between agents is modeled as an Erdős–Rényi (ER) graph with edge probability 0.5

Tables 10 and 11 report the results under iid data distributions. The results show that AdapAgg remains effective in the iid setting and achieves competitive or superior performance across the considered attacks. In particular, AdapAgg maintains high average test accuracy and strong minimum test accuracy among normal workers, indicating that the proposed adaptive aggregation mechanism is not limited to heterogeneous non-iid settings. These results empirically support Remark 3.

7 Conclusion

This paper introduced a novel parameter aggregation technique for peer-to-peer distributed learning that is resilient up to an arbitrary number of adversarial neighbors and applicable to non-convex loss functions and non-iid data distributions. The proposed aggregation rule leverages the similarity between workers' learning processes, forming an adaptive combination of neighbors' parameters based on these similarities. Through theoretical analysis, we demonstrated that the parameters of normal workers reach consensus when the underlying graph is connected. The parameters converge near the optimal value under the assumption of local strong convexity. Extensive empirical evaluations on standard machine learning tasks validate these findings, confirming the effectiveness of our method across diverse attack scenarios and varying numbers of adversarial workers.

The adaptability and robustness of the proposed approach make it suitable for a wide range of real-world distributed learning applications. Furthermore, since the method promotes similarity-based aggregation, it can be extended to other decentralized learning setups where encouraging agreement among participants is desirable. Extending this framework to

peer-to-peer learning with asynchronous communication between peers remains a promising direction for future research. Another prospective direction of future work is to extend the proposed adaptive aggregation method to dynamic communication graphs.

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Author Contributions C.B. developed the core methodology, implemented the algorithms, performed the theoretical analysis, and performed the experiments. C.B. and X.K. analyzed the results and interpreted the findings. C.B. wrote the main manuscript draft. X.K. provided supervision, conceptual guidance, and critical revisions. All authors reviewed and approved the final manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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